Exam 2 Review (up to 5 points extra credit)

Due to Gradescope by 1:00 p.m. on Wednesday, December 7th

You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade.

This extra credit opportunity cannot be accepted late.

Homework Goal: Practice the material we've covered for Exam 2 and bring questions and confusions to class on Wednesday, December 7th

NOTE: The exam will be on Chapters 7-9 and our extra materials after that. In addition to problems like the ones below, the exam will have true/false, fill in the blank, multiple choice, and matching. You would be wise to study all the terminology we have learned since Exam 1; a set of definitions are available on Knowt.io (link posted on Piazza). Also, homework solutions are posted on Canvas under Files.

Chapter 7: Networks

1. How long would it take to transmit a 10MiB file using each of the following media?
   (a) A 56 Kibps model (dial up)

$10\,MiB \rightarrow 80\,Mib \cdot 1024^2 = 83886080$

$83886080 / 56 \cdot 1024 = 1462.85\,s \rightarrow 24.38\,min$

   (b) A 1.5 Mibps DSL line (an older DSL line)

$$\frac{83886080}{1.5 \cdot 1024^2} = 53.33\,s \rightarrow 0.89\,min$$

   (c) A 100 Mibps Ethernet link (Fast Ethernet)

$$\frac{83886080}{100 \cdot 1024^2} = 0.8\,s$$

2. Name the layers in the Internet Protocol Stack and describe the goal of each layer.

Physical – Create a bit pipe between two computers

Data link layer – Create an error free bit pipe

Network layer – deliver a message from source to destination

Transport layer – Create a program-to-program delivery service

Application layer – provide rules to offer end-user services

3. On what layer is the Medium Access Control Protocol? According to this protocol, how must computers handle retransmission?

The MAC Protcol is on layer 2a the data link layer and computers must handle retransmission by using Automatic Repeat reQuest (ARQ).

4. On what layer are TCP and UDP used? What's the difference between TCP and UDP?

TCP & UDP are on layer 4 the transport layer. TCP is slower and more reliable it would be used for email while UDP is faster but less reliable used more for FPS gaming.

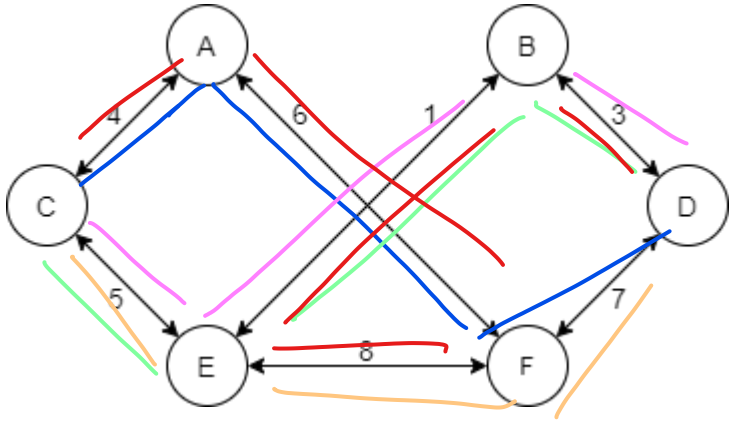
5. Suppose node A is sending node B a packet using the ARQ algorithm.
   (a) If packet 4 is lost on the way from A to B, which node(s) respond and how?

Node A won't recieve an ACK from Node B for packet 4 so A will time out and re-send packet 4. Once Node B recieves packet 4 it will save the packet and send Node A an ACK. Once Node A recievos the ACK it will delte its copy of packet 4.

   (b) If the ACK for packet 7 is lost on the way from B to A, which node(s) respond and how?

Node A won't recieve an Ack from Node B for Packet 7 and so node A will resend packet 7 to node B. Once Node B recieves its seccond copy of Packet 7 it will discard the copy and send another ACK back to Node A.

6. Consider the network shown below with the current measured delays between two nodes.



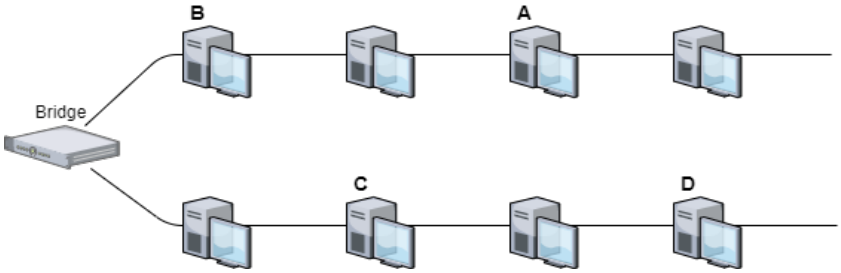
(a) List all possible simple paths between C and D; simple paths are those that do not repeat a node (i.e., no loops).

CAFD: 17
CAFEBD: 22
CEBD: 9
CEFD: 20

(b) Which path provides the shortest delay?

CEBD

7. The figure below shows two shared bus networks connected via a bridge.



(a) If A is transmitting a packet to B, will D look at the packet and ask "is this for me"? Briefly justify your response.

No because a bridge is a smarter device and knows which nodes are connected to it

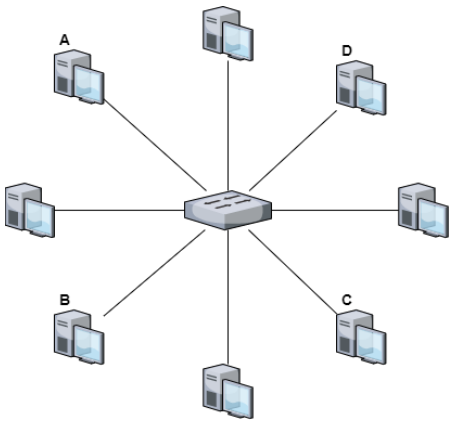
(b) If A is transmitting a packet to C, will D look at the packet and ask "is this for me"? Briefly justify your response.

Yes it will because the bridge will send. the message to all nodes connected to Node C

(c) Would your answers to part (a) or part (b) change if the bridge was a repeater instead? Explain your answer.

My answer to part A will change because a repeater repeats the message regardless of the destination node.

(d) Would your answer to a or b change if the network was a star with a central switch connected to all nodes as in the image below? Explain your answer.



Yes, a switch knows what nodes are connected to it so it would send the packet to correct Node and the others won't see the Packet.

Chapter 8/9: Privacy and Security

8. Explain the difference between a symmetric encryption algorithm and an asymmetric encryption algorithm. Then, define which type of algorithm each of the following are:
   (a) Caesar Cipher – symmetric
   (b) DES – symmetric
   (c) AES – symmetric
   (d) RSA – Asymmetric

   Difference:

   Symmetric : requires a secret key known to both sender and reciever; siegle key

   Asymmetric : requires a key for encryption and a different key for decryption; mulitple keys

9. Here is another simple hash algorithm, different from the one we discussed in class.

    1. Convert all characters to numbers (a = 1, …)
    2. Add 2 to every *number*
    3. Add together all the numbers
    4. Divide by 13 and keep the remainder
    5. Multiply the remainder by 19 and then add 5
    6. Reverse the digits
    7. Convert all digits to characters (a = 1, …)

    (a) My account is protected by the password “cherry”. Use the hash function to determine the hashed password.

    Cherry
    3 8 5 18 8 25
    5 + 10 + 7 + 20 + 20 + 27 = 89
    $\frac{89}{13} = 6.85$
    85 · 19 = 1615 + 5 = 1620
    0261 → abfa

    (b) Someone is attempting to log into my account by entering the password “cantaloupe”. Will they be granted access to my account? Explain why or why not and discuss why this might be possible for some attempts.

    cantaloupe
    3 1 14 20 1 12 15 21 16 5
    5 3 16 22 3 14 17 23 18 7
    $\frac{128}{13} = 9.85$
    85 · 19 = 1615 + 5 = 1620 → 0261

    Yes, they will be because when this password is hashed it equals the same hash as your password

10. Using a Caesar Cipher with shift = 18, encrypt the word “MANGO”. Then, by hand, decrypt the message “CDBKGLOBBI” and state the shift (may not be 18).

    ABCDEFGHIJKLMNOPQRSTUVWXYZ
    IJKLMNOPQRSTUVWXYZABCDEFGH

    MANGO → ESFYG

    ABCDEFGHIJKLMNOPQRSTUVWXYZ
    QRSTUVWXYZABCDEFGHIJKLMNOP

    CDBKGLOBBI → STRAWBERRY ; Shift 10

You've received a secret message from your friend detailing their favorite fruit! To keep it secret, they have encoded it using the block encoding scheme. For questions 17 and 18, use the encoding and decoding matrices provided below.

$$X = \begin{bmatrix} 4 & 3 \\ 9 & 7 \end{bmatrix} \qquad X' = \begin{bmatrix} 7 & 23 \\ 17 & 4 \end{bmatrix}$$

11. Decrypt the following message you received to figure out the fruit: IKZN

$S = \begin{bmatrix} 9 & 11 \\ 26 & 14 \end{bmatrix}$ $(X' \cdot S) \% 26 =$

I → 9
K → 11
Z → 26
N → 14

$\begin{bmatrix} 661 & 399 \\ 257 & 243 \end{bmatrix} \% 26 = \begin{bmatrix} 11 & 9 \\ 23 & 9 \end{bmatrix}$

KIWE

12. Reply to your friend by encoding the message: NEAT

N - 14
E - 5
A - 1
T - 20

$S = \begin{bmatrix} 14 & 5 \\ 1 & 20 \end{bmatrix}$

$(X \cdot S) \% 26 =$

$\begin{bmatrix} 59 & 80 \\ 133 & 185 \end{bmatrix} \% 26 = \begin{bmatrix} 7 & 2 \\ 3 & 3 \end{bmatrix}$

GBCC

13. Bob and Alice both have their own public key and private key. Bob wants to send Alice a message that only she can read and which she will know was written by him. How should Bob encrypt the message? How should Alice decrypt it?

Bob should encrypt the message first by signing it with private key then encrypt with Alice's public key

Alice should decrypt by first decrypting with her private key and then authenticate with Bob's public key

14. Suppose you and I both have a picture of a fruit basket, and this picture contains 1200 x 1800 RGB pixels. To send you a hidden message, I will reduce the intensity of each red, green, and blue value by one (and instead use those bits to hide my message).
    (a) How long of a message can I send you in bits?

$$12000 \cdot 18000 \cdot 3 = 6480000 \text{ bits}$$

    (b) If we use ASCII (8 bits), how many characters can I send you?

$$\frac{6480000 \text{ bits}}{8 \text{ bits}} = 810000 \text{ characters}$$

    (c) If we assume approximately 3,000 characters on a single-spaced page, how many pages of text can I hide in the image this way?

$$\frac{810000}{3,000} = 270 \text{ pages}$$

    (d) What is the technical term for hiding a message in this way?

Steganography

Data Science & Machine Learning

15. Suppose you want to investigate whether a person living near a garbage dump is more likely to get cancer. Briefly describe the four steps in this data science investigation and provide one task you would do in each step. Make your task as specific as possible.

The first step is obtain data I would do this by going to big data web sites and find cancer patients living conditions. After I would need to clean the data from all the garbage ; bolony that I do not need. from there I can explore the data and see how many patients that got cancer lived near a garbage dump. Finally, I try to predict the unknowns and see what other causes it could have been.

16. What is Big Data? Be specific.

Big Data has 3 V's: Volume, Velocity and Variety. There are other V's like Veracity which is accuracy of data and Variability which is the inconsistency of data. Big Data are huge data warehouses where the data can be used for analytical purposes. They also use the map reduce framework.

17. Briefly describe the difference between unsupervised and supervised machine learning. Use the terms *feature* and *label* in your description.

In supervised learning you teach machines by providing an example. It uses features which is the value that describes the data and labels which is the value being determined. However unsupervised learning is trying to recognize the patterns in data without examples. Those models just need to be provided features.

18. What is the difference between machine learning and deep learning? Use *neural networks* in your answer.

Machine learning is giving computers the ability to learn without being explicitly programmed. As for Deep Learning that is creating machine learning models that use deep neural networks.

AR/VR & Robotics

19. Name and give a brief description of the 3 D's of situations where robotics should be used in place of humans.

Dull - repetive, predictable, simple
Dirty - sanitation, sewer, repair
Dangerous - outer space, underwater, de-mining

20. Describe the difference between Video See Through and Optical See Through in AR.

optical see through is when the world is viewed directly through a lens or screen and virtual objects are rendered. Video see through is when the real world is viewed in directly through a video camera and virtual objects are rendered on the display.